Chapter 2: Asset Classes and Financial Instruments

"I'm a little concerned about my financial advisor. He said to keep 60% in equities, 25% in bonds, and 25% in cash."

A Tour of the Zoo

- This course covers three main financial instruments:
  - equity
  - fixed income
  - derivatives
- It's important to know (at least the names of) all the different sub-species of financial instruments.
- That's the goal of today's lesson.

The Money Market

- The money market is a subset of the bond market.
- These are short-term, highly liquid securities.
- They are often called *cash* or *cash equivalents*.

The Money Market

1. **Treasury Bills.**
   - This is the simplest money market instrument.
   - The US government sells bills to raise money.
   - Maturities of 4, 13, 26, and 52 weeks.
   - Not to be confused with Treasury Notes and Bonds.
   - Considered "risk-free".
2. **Certificates of Deposits.**
   - Certificates of Deposit (CDs) are time deposits with a bank
   - Unlike a checking account, you cannot withdraw funds on demand
   - They are insured by the FDIC and may be traded on a secondary market

The Money Market

3. **Commercial Paper.**
   - Large corporations issue short-term notes to finance working capital
   - They have maturities less than 270 days to avoid registration with the SEC
   - They were historically used by non-financial firms, but notoriously used by financial firms just before the crisis
4. **Bankers' Acceptances.**
   - Purchaser authorizes a bank to pay a seller for goods at later date (time draft)
   - When purchasers bank accepts draft, it becomes contingent liability of the bank (and marketable)
   - Often used to back international sales of goods and services

The Money Market

5. **Eurodollars.**
   - Dollar-denominated time deposits held outside U.S.
   - Pay higher interest rate than U.S. deposits due to liquidity risks
6. **Repos.**
   - Short-term sales of securities with promise to repurchase at higher price
   - RP is a collateralized loan
   - Many RPs are overnight; "Term" RPs may have a 1-month maturity
7. **Reverse Repos.**
   - Lending money; obtaining security title as collateral
   - "Haircuts" may be required depending on collateral

The Money Market

8. **Brokers' Calls.**
   - Call money rate applies for investors buying stock on margin
   - Loan may be "called in" by broker
9. **Federal Funds.**
   - Trading in reserves held at the Federal Reserve
   - Key interest rate for economy

The Money Market

10. LIBOR (London Interbank Offered Rate)
    - Second only to Treasuries in importance.
    - It is the rate at which banks lend to each other.
    - It is used as a benchmark rate for trillions of dollars worth of securities (primarily derivatives).

The Libor Scandal

My thwarted attempt to tell of Libor shenanigans

Douglas Keenan

In 1991, I began trading for Morgan Stanley, the investment bank, in London. I was trading bonds, derivatives and related securities. One of those securities was based on the three-month Libor rate: the interest rate at which banks can borrow money for three months from each other. Morgan Stanley does not trade on the interbank market so I could not directly borrow or loan money at Libor rates. What I could do, however, was trade a futures contract on the three-month Libor rate.

As an example of how a futures contract works, consider the following. Suppose that we are concerned about three-month Libor rates increasing in the future; in particular, we are concerned about what the three-month rate will be in September. If that rate is, say, 1 per cent, we can agree today to effectively lock it in. If, come September, the actual three-month rate is 2 per cent, then our contract will ensure we can still borrow at 1 per cent. Futures contracts on three-month Libor were – and are – traded on the London International Financial Futures Exchange (Liffe, now part of NYSE Euronext). There was a standard contract for the month of September. That contract had its rate settled on the third Wednesday of the month, at 11am.

In 1991, I had live trading screens that showed the Libor rates. In September of that year, on the third Wednesday, at 11am, I watched those screens to see where the futures contract *should* settle. Shortly afterwards, Liffe announced the contract settlement rate. Its rate was different from what had been shown on my screens, by a few hundredths of a per cent.

As a result, I lost money. The amount was insignificant for me, but I believed that I had been defrauded and I complained to Liffe. Liffe explained that the settlement rate was not determined by what rates were actually in the market. Instead, the British Banker's Association polled some banks, asking them what the rates were. The highest and lowest reported rates were discarded and the rest averaged, giving the settlement rate. Liffe explained that, in doing this, they were adhering to the terms of the contract.

I talked to some of my more experienced colleagues about this. They told me banks misreported the Libor rates in a way that would generally bring them profits. I had been unaware of that, as I was relatively new to financial trading. My naivety seemed to be humorous to my colleagues.

Simply put, then, it seems the misreporting of Libor rates may have been common practice since at least 1991. Although the difference between the reported rate and the actual rate might seem small, the total amount of money involved is material, given that Libor rates affect contracts worth hundreds of trillions. Also important is what such misreporting says about the culture.

I had been unaware of Libor misreporting, as I was new to trading. My naivety seemed humorous to colleagues

During 1991, at the London office of Morgan Stanley, the head of interest rate trading was a person who has been at the centre of the current scandal: Bob Diamond. I do not recall discussing Libor misreporting with Mr Diamond but since the misreporting was common knowledge among traders, I presume he was aware. (That, however, is not a criticism of Mr Diamond: what could he have done about this?)

There have been two distinct motivations for banks to misreport Libor rates. One motivation is discussed above: to directly increase profits. The other motivation arose during the 2008 financial crisis: to mask liquidity problems.

Libor misreporting has been going on for decades. Why have investigations only recently begun? It seems highly implausible that all the investigating agencies could have been unaware for decades. Indeed, those agencies have a reputation among traders of being like Potemkin villages. I suspect what has happened is that, after the financial crises of 2008, the agencies decided they ought to perform more of their stated duties. That would also explain why the investigations appear to be ignoring any misreporting in years before 2005: to cover up the illusoriness of their earlier work.

One of the investigations is being undertaken by the House of Commons Treasury committee. I telephoned the committee on July 3 and spoke with a committee specialist. I told the specialist about the foregoing and said that I was willing to testify under oath. The specialist seemed extremely interested. They said they were to have a meeting about the Libor scandal and would call me back afterwards. I did not hear back, however, so I phoned to ask what was happening. My testimony was not wanted, the specialist told me, because it "contradicts the narrative".

The writer is an independent mathematical scientist and a former Morgan Stanley trader

- Before 2012, LIBOR was literally a poll.
- Banks submitted what they "thought" they could borrow/lend at with other banks.
- It was discovered that banks were manipulating the poll to favor their trading positions.

The Bond Market

1. **Treasury Notes and Bills.**
   - Notes: up to 10 year maturities
   - Bills: up to 30 year maturities
2. **Treasury Inflation Protected Securities (TIPS).**
   - Principal adjusted for changes in the CPI
   - Marked with a trailing "i" in quote sheets

The Bond Market

3. **Federal Agency Debt.**
   - Most are home-mortgage-related: FNMA ("Fannie"), FHLMC ("Freddi"), GNMA ("Ginnie"), Federal Home Loan Banks
   - Risks of these securities?
     - Implied backing by the government
     - In September 2008, federal government took over Fannie and Freddie
4. **International Bonds.**
   - Bonds issued in another currency
   - E.g. Yankee bond is a bond sold in the US in USD by a non-US issuer; Samuari bond is a bond sold in Japan in Yen by a non-Japanese issuer, etc.

The Bond Market

5. **Municipal Bonds.**
   - Issued by state and local governments to finance capital improvement projects (e.g. roads, schools, parks).
   - Interest income is exempt from state and local taxation.
6. **Corporate Bonds.**
   - You know a lot about these already.
   - Two new species:
     - **Callable Bonds:** The firm can buy back the bond at a specified price.
     - **Convertible Bonds:** The bondholder can convert her bond into a specified number of shares of common stock.

The Bond Market

Call Me Maybe? The Effects of Exercising Contingent Capital

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Abstract

This paper studies market reaction and economic performance following the first episode of banks triggering contingent capital options. During the financial crisis, European banks massively used embedded options in their hybrid bonds to reduce their debt burden. These triggers are positively received by debtors, while stockholders discriminate according to the type of resulting debt relief and the financial institution leverage. Moreover, banks that trigger permanent debt reliefs exhibit higher economic performance than the ones that do not. These findings point towards innovative debt instruments offering an effective solution to the dilemma of bank capital regulation.

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The Bond Market

7. **Asset Backed Securities (ABSs).**
   - Put a bunch of assets in a portfolio called a "pool."
   - Create securities out of shares in the pool.
   - Voila! You've made an asset backed security.
   - Examples of assets turned into ABSs:
     - Mortgages ("Mortgage Backed Securities")
     - Life insurance policies ("death bonds")
     - Credit card debt
     - Student debt
     - Receivables
     - Home equity loans
     - Auto loans

Equities

1. **Common Stock.**
   - You know a lot about these already.
2. **Preferred Stock.**
   - Get paid before common stockholders, but can't vote
   - The firm has some discretion in paying preferred stock holders
3. **American Depositary Receipts.**
   - Certificates traded in the U.S. representing ownership in foreign security

Indices

- Indices are baskets of securities.
- They measure the return of a specific portfolio.
- Dow Jones Industrial Average (DIJA)
  - 30 large stocks
  - Price weighted average
- Standard & Poors Indices
  - S&P 500, S&P 400, S&P 600
  - Market value weighted average
- Specialized indices (e.g. FAANG)

Why do we have indices?

Indices

- Quick math review.
- Suppose we have N numbers, $p_1, p_1, \ldots, p_N$.
- We'll write

$$\sum_{j=1}^{N} p_j = p_1 + p_2 + \cdots + p_N \tag{1}$$

Price-Weighted (E.g. the DIJA)

- N stocks;
- p_j price of stock j;
- The *value* of the **price-weighted index** is

$$\frac{1}{N}\sum_{j=1}^{N} p_j \tag{2}$$

Example:

Company	**Price**
A	\$6
B	\$20

What's the *value* of the index?

Equally-Weighted

- N stocks;
- $p_{j,t}$ price of stock j on day t;
- The *return* of an **equally-weighted index** is

$$\frac{1}{N}\sum_{j=1}^{N}\frac{p_{j,t+1}-p_{j,t}}{p_{j,t}} \tag{3}$$

Example:

Company	**Price**$_t$	**Price**$_{t+1}$
A	\$4	\$3
B	\$10	\$11

What's the *return* of the index?

Indices

- Let's get fancy. Let's have N *weights* $w_1, w_2, \ldots, w_N$.
- The *weighted* sum of $p_1, p_1, \ldots, p_N$ is

$$\sum_{j=1}^{N} w_j p_j = w_1 p_1 + w_2 p_2 + \cdots + w_N p_N. \tag{4}$$

Example:

Company	**Weight**	**Price**
A	25	\$8
B	75	\$32

What's the weighted sum?

Value-Weighted (E.g. S&P 500)

- N stocks;
- $p_{j,t}$ price of stock j on day t;
- n_j number of shares of stock j.
- The *value* of the **value-weighted index** is

$$\sum_{j=1}^{N} n_j p_{j,t}. \tag{5}$$

- You can compute the return from day t to day $t+1$:

$$\frac{\sum_{j=1}^{N} n_j p_{j,t+1}}{\sum_{j=1}^{N} n_j p_{j,t}} - 1 \tag{6}$$

Value-Weighted (E.g. S&P 500)

Example:

Company	**Shares**	**Price**$_t$	**Price**$_{t+1}$
A	100	\$4	\$5
B	300	\$10	\$11

What's the return in the index from day t to day $t+1$?

Homework

Remember: 20 Questions on Chapter 2 (Nothing on Sec. 2.5)